INSTITUTIONAL COURSE SYLLABUS TEMPLATE

Course Code: **MATH 002**	Course Title: **Orientation Mathematics II**
Course Instructor: **Ms. Nada El Farra**	Email: **nelfarra@psu.edu.sa**
Credit Hours: **4**	Course Location: **Various**
Scheduled Office Hours: **Various**	
Office Location: **Various**	
Co-Requisite: **None**	Pre-Requisite: **MATH 001**

Mission Statement

The PYP is committed to achieving an effective transition of high school graduates into qualified vibrant learners who have the level of proficiency in academia necessary to participate fully as PSU students. Using the latest technological advances and scientific instruction, we seek to develop numerical, linguistic and study skills necessary to make the transition to college life. We are committed to delivering well-rounded individuals who have a passion for learning.

I. Course Description:

This is the second of the two-course sequence. It examines exponential and logarithmic functions, trigonometry, matrix algebra and conic sections needed for more specialized study in computer science, engineering and other related fields.

II. Course Learning Outcomes:

Skills	Course Learning Outcomes	Measured by
Knowledge	-Demonstrate an understanding and knowledge of Exponential, Logarithmic and Trigonometric functions and their graphs. -Apply basic features and operations on matrices and determinants.	Quizzes Online Practice Midterm Exam Final Examination
Skills	-Solve problems involving Exponential and Logarithmic functions and equations. -**Resolve** problems involving trigonometric functions, inverses and equations. - Solve systems of equations and inequalities using algebraic techniques, matrices and determinants. -Illustrate an understanding and knowledge of conic sections and their graphs	Quizzes Online Practice Midterm Exam Final Examination
Values		

III. Tentative Weekly Course Schedule:

Week	Date	UNIT/ TOPIC	CLO(s) alignment	STUDENT TASKS (quizzes& Midterm Exam)	STUDENT TASKS (Study Plan)	Contact hours
1	**Jan 14-Jan 18 Jan**	4.1 Exponential functions	1 3			4
2	**Jan 21-Jan 25**	4.2 Logarithmic functions 4.3 Properties of logarithms	1 3			4
3	**Jan 28-Feb 1** Jan 28 Extended weekend	4.3 Properties of logarithms 4.4 Exponential and logarithmic equations	1	Quiz 1 (4.1,4.2) Wednesday Jan 31		4
4	**Feb 4-Feb 8**	4.4 Exponential and logarithmic equations 5.1 Angles and their measure	3		Chapter 4 Feb 10	4
5	**Feb 11-Feb 15**	5.2 Right triangle trigonometry 5.3 Trigonometric functions of any angle	1 4	Quiz 2 (4.3,4.4) Wednesday Feb 14		3
6	**Feb 18-Feb 22** Feb 22 Saudi Founding day	5.3 Trigonometric functions of any angle 5.5 Graphs of sine and cosine functions	1			4
7	**Feb 25-Feb 29**	5.5 Graphs of sine and cosine functions 5.7 Inverse trigonometric functions	1 4	Quiz 3 (5.1,5.2) Wednesday Feb 28		4
8	**March 3-March 7**	6.2 Sum and difference formulas 6.5 Trigonometric equations	4 5	Midterm Exam Chapter 4 & Chapter 5 Thursday March 7		4
9	**March 10-March 14**	6.5 Trigonometric equations 8.1 Systems of linear equations in two variables			Chapter 5&6 March 16	4
Eid Al-Fitr Holiday						
10	**April 14- April 18** April 15 Classes resume	8.2 Systems of linear equations in three variables 8.5 Systems of inequalities	5			4
11	**April 21-April 25**	9.1 Matrix solutions of linear systems 9.2 Inconsistent and dependent System	5	Quiz 4 (6.2,6.5 & 8.1) Wednesday April 24	Chapter 8 April 27	4
12	**April 28-May 2** May 2 Extended weekend	9.3 Matrix operations and their applications 9.4 Matrix inverses and matrix equations	2 5			4
13	**May 5-May 9**	9.4 Matrix inverses and matrix equations 9.5 Determinants and Cramer's Rule	2 5	Quiz 5 (8.2,8.5,9.1&9.2) Wednesday May 8	Chapter 9 May 11	4
14	**May 12-May 16**	10.1 The Ellipse 10.2 The Hyperbola	2 5			4
15	**May 19-May 23**	10.3 The Parabola	6			4
16		Final Exam				

Omitted Objectives (Topics) and Examples

Section	Objective	Example		Section	Objective	Example		Section	Objective	Example
4.1	4	6		8.1	5	--		9.4	3	--
4.4	--	4 , 10		8.2	3	--		9.5	5	--
5.1	8	--		8.5	3	--		10.1	3 , 4	--
5.5	5 , 6	--		9.1	--	4		10.2	4 , 5	--
6.2	3	3 , 6		9.2	3	--		10.3	3 , 4	--
6.5	5	--		9.3	7	--				

IV. Student Assessment & Teaching Strategies: Assessment Task

Domain	Assessment Task	Teaching Strategy
Knowledge	**Quizzes, Online Practice, Midterm Exam, and Final Exam**	**Lectures, Discussions, Examples & Tutorials**
Skills	**Quizzes, Online Practice, Midterm Exam, and Final Exam**	**Lectures, Discussions, Examples & Tutorials**
Values		

V. Course Requirements

Quizzes, Online Practice, Midterm Exam, and Final Exam

VI. Schedule of Assessment

Assessment	Assessment Task	Week Due	Proportion of Final Assessment
1	Quizzes (5 Quizzes)	(shown in Syllabus-Pages 2&3)	30%
2	Midterm Exam	Thursday March 7	20%
3	Online Lab practice (Study Plan)	(shown in Syllabus-Pages 2&3)	10%
4	Final Examination	Week # 16	40%

VII. Learning Resources

A. References

Required Textbook:

Algebra and Trigonometry, Global Edition Seventh Edition, Prentice Hall, 2018
Robert Blitzer

Recommended Textbooks:

1. Algebra and Trigonometry, Sixth Edition, Prentice Hall, 2018
   Robert Blitzer

2. Algebra and Trigonometry, Fourth Edition; James Stewart; Cengage Learning, 2016.

3. College Algebra and Trigonometry, 7th edition
   By Richard N. Aufmann, Vernon C. Barker, and Richard D. Nation

4. College Algebra with Trigonometry, 9th edition
   By Raymond A. Barnett, Michael R. Ziegler, Karl E. Byleen, Dave Sobecki

Electronic Materials, Web Sites

MyLab Math, access to University LMS

B. Facilities Required

a. Classrooms with capacity of 30 students
b. Whiteboard
c. Data show projector and screen
d. Smart board and online course.

C. EDU-HUB Platform

LMS is efficiently used, Lecture Notes and Important instructions can be found on the university LMS at https://lms.psu.edu.sa/

VIII. Classroom Policies

A. Academic Integrity Policy (e.g. plagiarism or dishonesty)

Plagiarism can be defined as unintentionally or deliberately using another person's writing or ideas as though they are one's own. Plagiarism includes, but is not limited to, copying another individual's work and taking credit for it, paraphrasing information from a source without proper documentation, and mixing one's own words with those of another author without attribution. In addition, buying a paper or project, or downloading a paper from the Internet, and submitting them as your own are also plagiarism. The penalty for academic dishonesty will bring course expulsion and failure, or even suspension.

B. Attendance Policy

- Students are required to attend all classes starting from **the first day of the semester**.

- Attendance will be taken at the start of the lecture. A student will be marked **late** if he/she enters the classroom **after 5 minutes**. Every **3 lates** will be counted as **an absence**.
  If the student enters the classroom **after 10 minutes,** he/she will be marked absent.

- Students will be notified via text messages on their mobiles when they reach **1st warning (6 absences) and 2nd warning (11 absences). An automatic** "**DN Grade**" will be issued to a student when he/she reaches **16 absences** and a text message will be sent to his/her mobile. This means he/she cannot enter any more classes or exams.

- ***It is the sole responsibility of the student to keep satisfying level of attendance. Otherwise, "DN" grade will be granted automatically during anytime in the semester without any notice.***

C. Quizzes

5 quizzes will be given during the semester. **Each quiz will be worth 6% of the total grade.** The dates and the materials of the quizzes are shown in the "Weekly Course Schedule" above.

All 5 quizzes will be counted, and that requires continuous preparation by the student which is needed to be ready for the quizzes, **Mid-term Exam** and therefore for the **Final Exam**. The student should make sure that he/she is well prepared for every quiz.

Absence for Quizzes and Exams

- If you are absent for a quiz or the Midterm, then you will be given a zero unless you submit a valid excuse according to the University Rules and Regulations. Check with the students Affairs and/or your teacher for the Rules and Regulations on **"Acceptable and Non-acceptable Excuses"**

  Acceptable Excuses
  - Car accident with a detailed **Najm report** mentioning the driver's information, and date/time of the accident.
  - Hospital admission in **a recognized hospital (Government or private)** with a medical report signed by the hospital Director.
  - Death of a close **(immediate relative-first degree)** family member. The student should submit a copy of the Death Certificate to Student Affairs.
  - Caring for a sick family member in an emergency. A justification letter must be provided.
  - **One-day sick leave from a recognized hospital** with the reason clearly mentioned.
  - Force majeure – Credible evidence necessary

  Note: All justifications and excuses must be submitted within 10 days of the exam/assessment/quiz/assignment date.

  Unacceptable Excuses
  - **Excuses from medical centers, and small clinics**.
  - Oversleeping
  - Tiredness
  - Bad weather
  - Alarm clock not working - Not sure about the exam time
  - Traffic jam
  - Traveling or changing a flight schedule
  - Medical appointment
  - Migraine, fever, headache,...etc.
  - Commuting
  - Attending ceremonies: wedding, meeting, activities...etc.
  - Family issues
  - Dental emergencies

- If you are absent for the Final Exam, **you need to provide a legitimate excuse (according to PSU Rules and Regulations). Your case will then be reviewed by the University Council which will make a decision after reviewing your excuse.**

D. Online Work (Study Plan Practice)

- There is an online tool "**MyLabMath**" which is a very helpful tool that should improve the student's performance in the course with all the helping features available that should be utilized effectively.

- An **Access Code** for registering for the online work is needed. Information about buying the access code and registering for it (or enrolling in the new course for the student that already has an account), will be provided by your instructor.

- MyLabMath should be used regularly for practice throughout the semester using "**Study Plan**".

- The grade assigned to the "Online Practice" (10%) will be granted to the student based on the number of problems she practices using the **"Study Plan**". The student's online practice will be monitored and assessed regularly throughout the semester and therefore the grade will be granted based on that.

- A student is required to do **at least 300 practice problems (Correct).** Use the following table as a guide.

	Chapter	No. of correct questions	Weight out of 10%	Due date
Study plan#1	4	75	2.5	Midnight of Saturday, Feb 10
Study plan#2	5&6	100	3	Midnight of Saturday, March 16
Study plan#3	8	50	2	Midnight of Saturday, April 27
Study plan#4	9	75	2.5	Midnight of Saturday, May 11

- Please note that the study plan **will not** be opened for any reason whatsoever after the deadline of the study plan has passed.

Calculators:

- Only Scientific Calculators are allowed in this course (No Graphing Calculators).

Office Hours:

- You are advised and encouraged to seek help to clarify matters that are not clear to you as soon as possible.
- Check the table posted on your instructor's door for the office hours in case you need assistance or you need to inquire about matters concerning your marks, absence, and so on. If you need to see her/him at a different time, arrange with her/him in advance.

Online Practice-1-)2.5%)

(Chapter# 4)

Due: The midnight of Saturday, Feb 10

# of correct practice problems	Mark
1-17	0.5
19-37	1
38-56	1.5
57-74	2
75 or more	2.5

Online Practice-2-(3%)

(Chapter#5&6)

Due: The midnight of Saturday, March 16

# of correct practice problems	Mark
1-19	0.5
20-39	1
40-59	1.5
60-79	2
80-99	2.5
100 or more	3

Online Practice-3-(2%)

(Chapter# 8)

Due: The midnight of Saturday, April 27

# of correct practice problems	Mark
1-13	0.5
14-31	1
32-49	1.5
50 or more	2

Online Practice-4-(2.5%)

(Chapter# 9)

Due: The midnight of Saturday, May 11

# of correct practice problems	Mark
1-17	0.5
19-37	1
38-56	1.5
57-74	2
75 or more	2.5